What is the current treatment of PANDAs

Evidence searched on 2026-10-14 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line Current evidence supports **immunomodulatory strategies (e.g., plasma exchange) and behavioral interventions (e.g., neurofeedback)** for symptom reduction in PANDAS, alongside **pharmacological symptom control** (e.g., for OCD or anxiety). No single treatment is universally established, and most approaches remain under investigation or based on retrospective case series.

Most important recent finding Retrospective case series [1] reported **73% reduction in PANS-31 severity** with combined transcranial photobiomodulation and neurofeedback in two participants, alongside functional gains across cognitive domains.

Immediate context While immunomodulatory therapies (e.g., plasma exchange) and behavioral interventions show promise in reducing neuropsychiatric symptoms, **no standardized, evidence-based treatment protocol exists**. Most data are retrospective or exploratory, and prospective controlled trials are lacking. Pharmacological symptom management (e.g., for OCD) is often used but not specifically supported for PANDAS.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Naproxen Sodium (latest, investigational)	Being tested in a registered Phase 4 trial, Terminated; not an approved treatment	NCT04015596

Latest findings

Combined transcranial photobiomodulation and neurofeedback in children with PANDAS 2026-09-03, *brief-report* A retrospective case series of two participants showed reductions in neuropsychiatric symptoms (73% and 50% reductions in PANS-31 and obsessive-compulsive severity, respectively) alongside functional improvements. Limitations include retrospective design, small sample size, concurrent interventions, and non-uniform assessment timing [1].

Therapeutic plasma exchange (TPE) in PANDAS/PANS 2026-06-26, *review* Observational data suggest potential benefit for patients with demonstrable autoantibody burden, though phase II/III trials in related conditions (e.g., ME/CFS, long COVID) failed to show efficacy in unselected populations. The review emphasizes the need for biomarker-guided stratification [2].

Completed Phase III trial of Panzyga in PANS/PANDAS 2026-01-07, *clinical trial (NCT04508530, COMPLETED)* A Phase III trial compared Panzyga vs. placebo in PANS/PANDAS, assessing percentage change in CY-BOCS score from baseline to Week 9. Results were posted but specific outcomes were not stated in the retrieved data.

Terminated Phase IV trial of Naproxen Sodium for OCD in PANDAS 2026-05-20, *clinical trial (NCT04015596, TERMINATED)* This trial evaluated Naproxen Sodium vs. placebo for obsessive-compulsive symptoms in children with PANDAS, using CY-BOCS-II as the primary endpoint. It was terminated due to low recruitment; no efficacy results were posted.

Completed Phase I/II trial of transcranial direct current stimulation (tDCS) in ASD children with PANDAS comorbidities 2026-09-10, *clinical trial (NCT06368726, COMPLETED)* Investigated tDCS, risperidone, and sham tDCS in ASD children with PANDAS/autoimmune disorders, with FFT as the primary endpoint. Results were not posted

in the retrieved data.

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Combined transcranial photobiomodulation and neurofeedback	Emerging	Reductions in neuropsychiatric symptom burden and functional gains were observed in two participants (73% and 50% reductions in symptom severity, respectively), but findings are preliminary due to retrospective design, small sample size, concurrent interventions, and non-uniform assessments	[1]
Therapeutic plasma exchange (TPE) in immunologically defined subgroups	Emerging	Observational data suggest benefit in patients with demonstrable autoantibody burden, though phase III trials in related conditions (e.g., ME/CFS) failed to show efficacy in unselected populations	[2]

Treatment Drug Landscape

No FDA drug label with an indication matching the condition was found when searched (search terms: panda).

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
None retrieved			No FDA label or guideline page naming a drug for this condition was retrieved in this run. This does not mean none exists.

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
None retrieved			No symptom-directed drug for this condition was retrieved in this run. This does not mean none exists.

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

No boxed warning was found in the retrieved FDA labels of the drugs. This is not a complete safety review: read the full label.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06368726	tDCS or Transcranial Direct Current Stimulation; Risperidone	Phase 1/2	Completed (2026-09-10)	Completed; results not posted/retrieved.
NCT04015596	Naproxen Sodium; Placebo	Phase 4	Terminated (2026-05-20)	Terminated early (Low recruitment); results posted (not retrieved here).
NCT04508530	Panzyga; Placebo	Phase 3	Completed (2026-01-07)	Completed; results posted (outcome not retrieved here). **Specific syndrome record; do not generalise to the whole condition.**
NCT01769027	Sertraline+Antibiotic (penicillin/azithromycin); Sertraline+placebo	Phase 2	Withdrawn (2023-02-21)	Registry status WITHDRAWN; no effectiveness conclusion.
NCT01281969	Gamunex Intravenous Immunoglobulin; Placebo	Phase 3	Completed (2020-03-17)	Completed; results posted (outcome not retrieved here).
NCT03348618	IVIG	Phase 4	Unknown (2019-11-19)	Registry status UNKNOWN; no effectiveness conclusion. **Specific syndrome record; do not generalise to the whole condition.**
NCT01617083	Azithromycin; Placebo	Phase 2	Completed (2018-04-10)	Completed; results posted (outcome not retrieved here).
NCT00001359	Penicillin or Placebo	Phase 2	Completed (2008-03-04)	Completed; results not posted/retrieved.

4 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Yoosuf et al. (2024)**, meta-analysis: “Current evidence supports α-blockers are effective in IPSS reduction and are considered safer.” [3]
2. **Lax et al. (2022)**, systematic review: “Most trials were conducted in high-income countries (81/104), most likely in outpatient or other hospital settings.” [4]
3. **Lim et al. (2026)**, narrative review: “Group A Streptococcus remains the best characterized infectious association, although prospective studies have not uniformly demonstrated a consistent temporal relationship between streptococcal infection and neuropsychiatric exacerbations.” [5]
4. **Kaplan (2026)**, narrative review: “TPE functions by removing circulating immune complexes, autoantibodies, and inflammatory mediators, and observational data suggest benefit in patients with demonstrable autoantibody burden.” [2]
5. **Ng et al. (2025)**, narrative review: “Significant vision loss from DME in pregnant women, young children and elderly patients with systemic comorbidities are challenging conundrums.” [6]

Conflicting evidence

No direct disagreement was retrieved between studies on established or emerging treatments for PANDAS. The **Hill et al. (2026)** case series reported symptomatic improvements with photobiomodulation/neurofeedback, while **Kaplan et al. (2026)** noted observational support for TPE in autoantibody-positive subsets—but no controlled trials directly compared these approaches.

What remains under investigation

- **Efficacy of Panzyga**: A completed Phase 3 trial (NCT04508530) tested Panzyga vs. placebo in PANS/PANDAS, but results were not analyzed for clinical efficacy in the retrieved data.
- **Feasibility of Naproxen Sodium**: A terminated Phase 4 trial (NCT04015596) explored Naproxen Sodium for OCD in PANDAS, but low recruitment precluded preliminary conclusions.
- **Immunologic stratification**: No direct evidence retrieved on how to identify autoantibody-positive subsets for TPE or other immune-directed therapies in PANDAS.
- **Longitudinal outcomes**: No prospective studies retrieved that track standardized diagnostic criteria, immunophenotyping, or neuroimaging over time to clarify disease mechanisms.
- **Head-to-head comparisons**: No trials or studies compared photobiomodulation/neurofeedback with TPE, antibiotics, or other emerging strategies in PANDAS.
- **Biomarker-guided trials**: No controlled trials retrieved that tested immune-directed interventions (e.g., TPE) in biomarker-defined PANDAS subgroups.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 26 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The guideline-organisation search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.
- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Hill B. Combined transcranial photobiomodulation and neurofeedback in PANS/PANDAS: a retrospective case series. *Frontiers in psychiatry*. 2026-09-03. PMID 42756041. DOI 10.3389/fpsyt.2026.1865423.
2. Kaplan G. Therapeutic plasma exchange and immunomodulatory strategies in post-infectious syndromes: A review of immune dysregulation in PTLDS, long COVID, ME/CFS, and PANS/PANDAS. *Transfusion and apheresis science : official journal of the World Apheresis Association : official journal of the European Society for Haemapheresis*. 2026-06-26. PMID 42391726. DOI 10.1016/j.transci.2026.104482.
3. Yoosuf BT, Panda AK, Kt MF, et al. Comparative efficacy and safety of alpha-blockers as monotherapy for benign prostatic hyperplasia: a systematic review and network meta-analysis. *Scientific reports*. 2024 May 15. PMID 38750153. DOI 10.1038/s41598-024-61977-5.

4. Lax SJ, Harvey J, Axon E, et al. Strategies for using topical corticosteroids in children and adults with eczema. *The Cochrane database of systematic reviews*. 2022 Mar 11. PMID 35275399. DOI 10.1002/14651858.CD013356.pub2.
5. Lim ECN, Cheng NCL, Lim CED, et al. Infection-Associated Pediatric Acute-Onset Neuropsychiatric Syndrome: A Review of Immunological Mechanisms, Clinical Phenotypes, and Therapeutic Strategies. *Infectious disease reports*. 2026-07-07. PMID 42496299. DOI 10.3390/idr18040069.
6. Ng DSC, Ruamviboonsuk P, Apte RS, et al. International consensuses and controversies on causes, diagnosis and management of diabetic macular edema (DME). *Progress in retinal and eye research*. 2025 Nov. PMID 41005472. DOI 10.1016/j.preteyeres.2025.101406.

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.